Hemanth Sridhar Nakshatri

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Education

University of Wisconsin — Madison *Madison, WI*
Master of Science in Electrical and Computer Engineering (Machine Learning & Signal Processing) *Sept 2023 – Dec 2024*
- **Coursework:** Machine Learning (760), Big Data Systems (544), High-Performance Computing (759), Computer Vision (766), Computer Architecture (552), Image Processing (533)

Dayananda Sagar College of Engineering *Bangalore, India*
Bachelor of Engineering in Electronics and Communication Engineering *Aug 2017 – May 2021*

Experience

Cloud Engineer II / Software Engineer (Systems) *San Jose, CA*
Hewlett Packard Enterprise *Feb 2025 – present*
- Architected an **enterprise multi-agent AI harness** for automated Root Cause Analysis (RCA) on HPE's observability platform (OpsRamp), orchestrating specialized agents across alerts, traces, logs, and topology for enterprise ITOM.
- Engineered a **600+ tool orchestration layer** with dynamic selection and context-aware routing, eliminating context-window bloat while maintaining full coverage — backed by **Kafka, Redis, Cassandra, and Elasticsearch**.
- Architected an **extensible, MCP-compatible plugin system** (Bring Your Own Skills, Rules, and Evidence), enabling tenants to integrate external telemetry sources alongside native HPE capabilities.
- Built a **continuous learning engine** that retains past investigation outcomes, learns from corrective feedback, and adapts to user preferences over successive triage cycles.
- Developed **agentic platform control and onboarding automation**, enabling agents to autonomously navigate multi-page telemetry and condensing multi-hour configuration workflows into minutes.
- Slashed false-positive escalation to on-call engineers by over **80%**, accelerating Mean Time to Diagnosis (MTTD) to under **10 minutes** across multi-domain hypotheses.

Software Engineer Intern *San Jose, CA*
Hewlett Packard Enterprise *June 2024 – Aug 2024*
- Built an end-to-end RAG virtual assistant utilizing dynamic context distillation inspired by Map-Reduce, enabling LLMs to accurately synthesize 100k+ word knowledge bases.
- Designed a scalable architectural pipeline for time-series forecasting, capable of ingesting and modeling diverse univariate and multivariate incoming telemetry streams.

Project Associate *Bangalore, India*
Indian Institute of Science *Aug 2021 – July 2023*
Supervised by Prof. Jaya Prakash
- Formulated model resolution matrix-based deconvolution algorithms for computational medical imaging, boosting reconstruction quality in photoacoustic tomography by over **20% in PSNR** and **30% in SSIM**.
- Developed a novel complex-valued LSTM neural network architecture for frequency-domain imaging, outperforming baseline methods by **15% in PSNR**, **10% in SSIM**, with a **75-fold speedup**.

Patents and Publications

- **Nakshatri, Hemanth S, et al.** Synchronized Context Management System for AI in IT Operations Management Applications. (**U.S. Patent Pending**)
- Munjal, Isha, **Nakshatri, Hemanth S**, Jaya Prakash. "Low dose computed tomographic image reconstruction with advanced multi-encoder architecture". ***Journal of Instrumentation (2025)***
- **Nakshatri, Hemanth S**, Jaya Prakash. "Complex Valued Long Short Term Memory Based Architecture for Frequency Domain Photoacoustic Imaging". ***IEEE Transactions on Computational Imaging (2023)***
- **Nakshatri, Hemanth S**, Jaya Prakash. "Model resolution matrix based deconvolution improves over non-quadratic penalization in frequency-domain photoacoustic tomography". ***The Journal of the Acoustical Society of America (2022)***

Technical Skills

Programming: Python, Java, C, C++, Verilog, MATLAB, SQL

Generative AI & Agents: Multi-Agent Systems, Model Context Protocol (MCP), A2A, LangChain, LangGraph, Google ADK, RAG, Prompt Engineering

ML & Signal Processing: PyTorch, TensorFlow, Scikit-learn, CUDA, Computer Vision, Signal Processing, Time-Series Forecasting

Systems & Cloud: Kafka, Redis, Cassandra, Elasticsearch, Docker, Kubernetes, Spark, Spring, J2EE, Linux